1. Run report: lit2026 W10, run-20261011T2118Z

What ran. `examples/lit2026_w10_coppersmith_bound.rs` at commit `a21b5de50` (branch `lit2026/w10-s5-coppersmith`, stacked on the program commit `3c239129d`), built with `CARGO_TARGET_DIR=/Volumes/SSD990-2/crypto/targets/lit2026-w10 cargo build --release --example lit2026_w10_coppersmith_bound`, run as `./lit2026_w10_coppersmith_bound --part all --max-ell 3 --json run.json` (wrapper `run_w10.sh`). Start 2026-10-11 21:18 UTC, end 22:01 UTC; 2564 s wall, 301 s user, peak RSS 1.45 GB (`time_and_exit.txt`). Host: shared Apple M4 Pro, load average 108 at start, 177 at end (`host_load.txt`). Exit status 0. No run failed or timed out.

Isolation. None claimed. Every reported quantity except the `ms` columns is a deterministic count or an exact rational (closure optimum, monomial counts, LLL vanishing counts with fixed seeds). Wall times are L0, single sample, informational only.

Raw records. `run.md` (all tables), `run.json` (schema `lit2026-w10-coppersmith/v1`, 181 records). Graphs are drawn from `run.json` by `../../plot_w10.py` (layout only).

1.1. Parts and what each measured

1. **Known bounds** (part `known`). Univariate degree $d \in \{2, 3, 5\}$, $\ell \le 16$: exact closure optimum equals $\frac{\ell}{d\ell+d-1}$ in all 30 cells and tends to $\frac{1}{d}$. Boneh–Durfee $f = 1 + x(A + y) \bmod e$, $X = e^{\delta}$, $Y = e^{\frac{1}{2}}$: the exact closure optimum equals the Boneh–Durfee x+y-shift lattice with the best t at every $\ell \in \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32\}$ (0.27949 at $\ell = 32$); the closed form at $\ell = 2000$, $\tau = 0.215$ gives 0.28466 against the literature value $\frac{7-2\sqrt{7}}{6} = 0.28475$. Jochemsz–May basic lattice on the box $[0, d]^m$: 1/18, 1/10, 0.11905 at $\ell = 1, 2, 3$ for S_3 (the repo's probe-B column) and limits $\frac{1}{6}, \frac{1}{24}, \frac{1}{80}$.
2. **Supports** (part `support`, 61-bit prime, random a, b, random target z). S_3, S_4, S_5 with $x_{m+1} = z$ have 9/9, 125/125, 6561/6561 monomials of the box $\left[0, 2^{m-1}\right]^m$, all box vertices, total degree 4, 12, 32; symmetric; the interpolant equals the direct resultant at 8/8 off-grid points, Semaev vanishes at 8/8 random relations and the interpolant at 4/4 planted decompositions of z. Elementary-symmetric rewriting: 6, 35, 495 monomials, weighted degree 4, 12, 32.
3. **Box families** (part `families`): JM basic and best-extension box for $\ell \le 128$.
4. **Closure optimum** (part `closure`): exact maximum-weight-closure optimum over all monomial sets in $[0, L]^m$ for each leading vertex class.
5. **Finite lattices** (part `lattice`): 32-bit curve `a3-fast-32bit-p256class` ($p = 4294967291$, $a = -3$, $b = 31$, prime order $n = 4294959973$), repo `lll_reduce` (≤ 30 dims) / `lll_reduce_hp`, Lovász 0.99.

1.2. Closure optimum (selected rows; all rows in `run.md` §4)

poly	ℓ	vertex classes	L	δ_ℓ exact	δ_ℓ	closure = best box $\le L$	dim
S_3	1	0,1,2	4	9/100	0.090000	yes	25
S_3	12	0,1,2	48	202/1305	0.154789	yes (L′=29)	900
S_4	1	0..3	18	1331/70875	0.018780	yes (L′=14)	3375
S_4	5	0..3	41	25090/761349	0.032955	yes (L′=37)	54872
S_5	1	0,4	44	923521/175821516	0.005253	yes (L′=38)	2313441
S_5	1	0..4	24	83521/18750000	0.004454	yes (bounded)	390625
S_5	2	0,4	32	237073/37949472	0.006247	yes (bounded)	1185921
S_5	2	1..4	24	45041/9375000	0.004804	yes (bounded)	390625
S_5	3	4	32	480707/75898944	0.006334	yes (bounded)	1185921

In every one of the 70 closure cells the optimum equals the best box $[0, L']^m$, $L' \le L$, and every vertex class gives the same value. The charging check $\Sigma k \le \Sigma_{u \in M} J(u)$ held in all cells (with equality). "Bounded" means the best box is the bounding box itself.

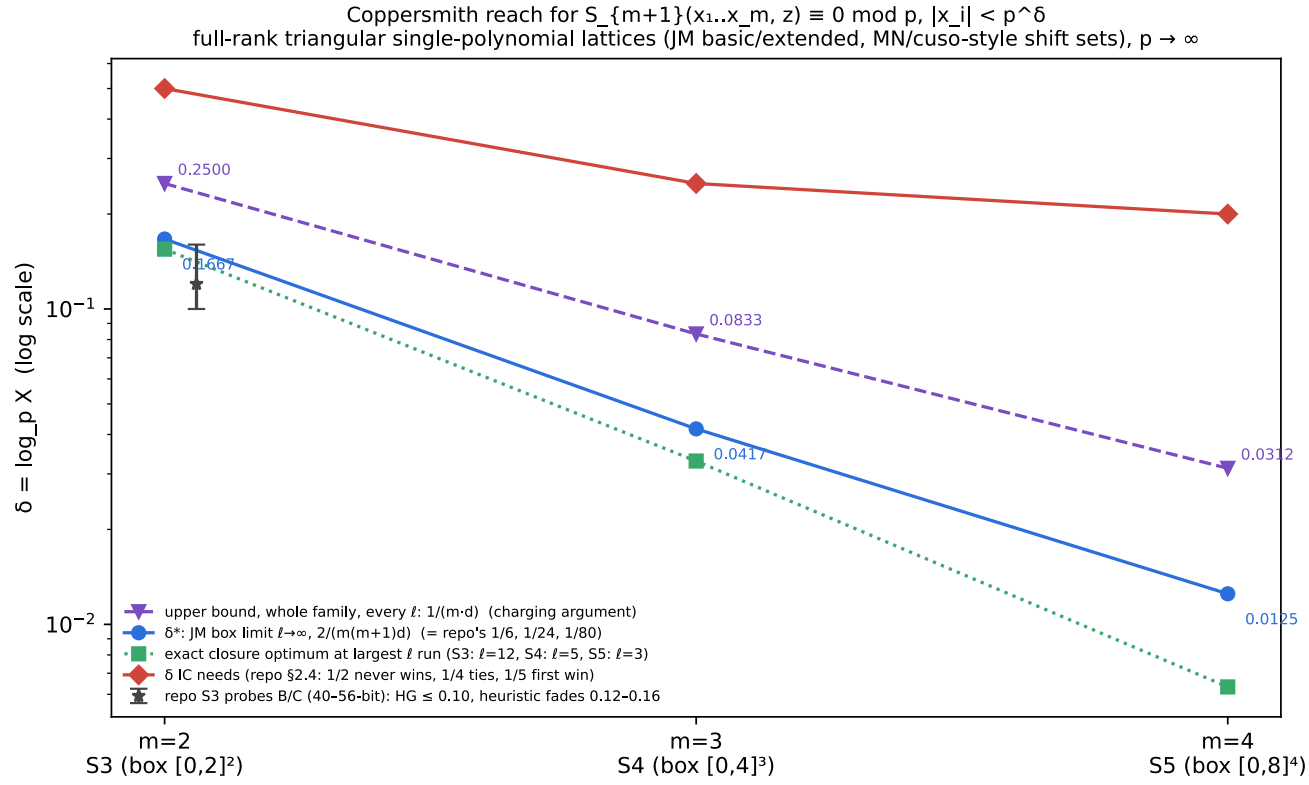
Symmetric encoding $G(e_1, \ldots, e_m)$ with weights $i\delta$ (§4b): S_3 0.100000, 0.125000, 0.136111 at $\ell = 1, 2, 3$; S_4 0.018176, 0.024903 at $\ell = 1, 2$; S_5 0.004061 at $\ell = 1$ (bounding box $[0, 2 \cdot \max \deg]$; best vertex the constant term).

1.3. Finite lattices

poly	ℓ	dim	δ (B)	$\ge m$ rows vanish	HG first m rows	mean vanishing
S_3	1	9	0.10 (9)	6/6	3/6	2.2
S_3	1	9	0.14 (22)	3/8	0/8	1.5
S_3	1	9	0.16 (34)	0/7	0/7	0.4
S_3	2	25	0.12 (14)	7/7	1/7	12.1
S_3	2	25	0.16 (34)	4/8	0/8	1.4
S_3	2	25	0.20 (84)	0/8	0/8	0.1

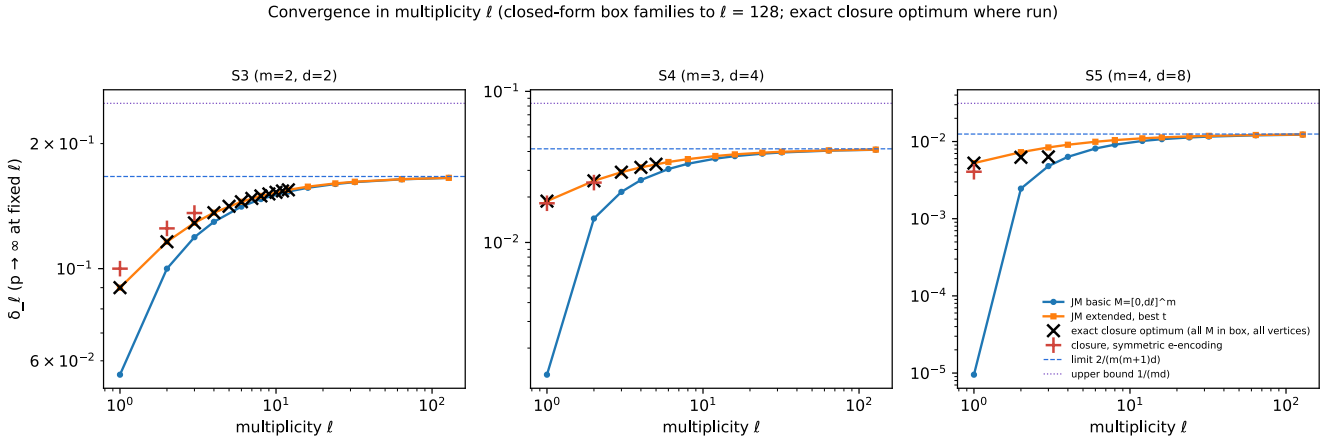
S_4	1	125	0.035 (2) … 0.125 (15)	0/36	0/36	0.0–0.7

S_4 at $\ell = 1$ recovered no planted root at any tested bound, including the smallest ($B = 2$); its asymptotic-in-p reach at $\ell = 1$ is $\delta_1 = 0.00133$, i.e. $X < 1.03$ at 32 bits. Trials per cell: 3–8 for S_3 (a trial is dropped, not counted, when the planted sum is the point at infinity — frequent at $B \leq 3$ — or LLL returns an error), 6 for S_4.



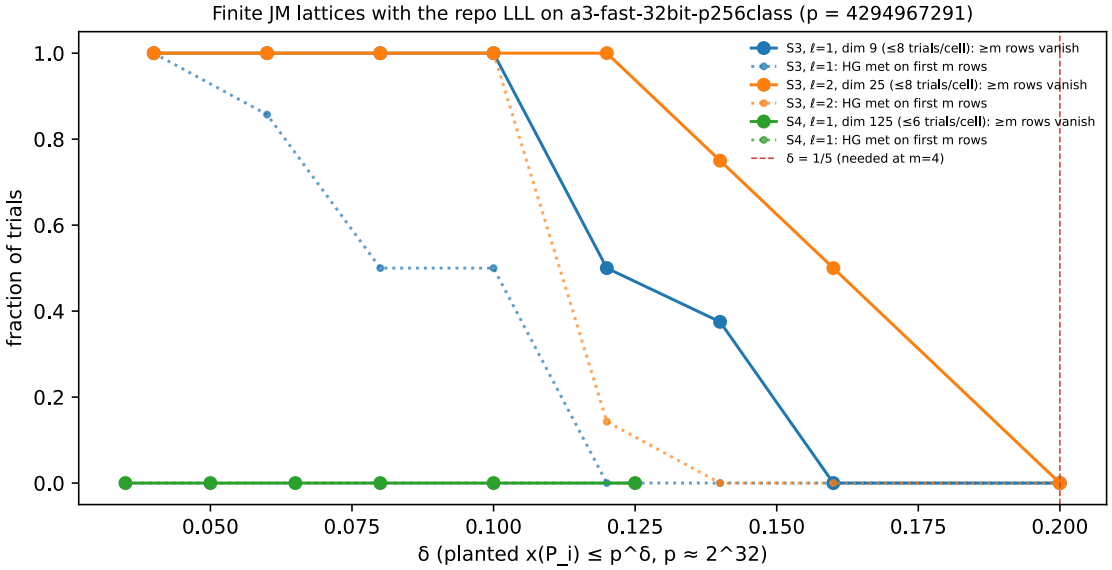
Source: research/lit2026_w10_s5_coppersmith_20261011/reports/run-20261011T2118Z/run.json (lit2026 W10, examples/lit2026_w10_coppersmith_bound.rs)

Figure 1: δ against arity: upper bound, JM limit, closure optimum, needed δ, repo S3 probes. Source `run.json`.



Source: research/lit2026_w10_s5_coppersmith_20261011/reports/run-20261011T2118Z/run.json

Figure 2: δ_ℓ against multiplicity ℓ. Source `run.json`.



Source: research/lit2026_w10_s5_coppersmith_20261011/reports/run-20261011T2118Z/run.json

Figure 3: Finite lattices on the 32-bit curve. Source `run.json`.